Rashik Intisar Siddiquee

3533 Ivy Commons Drive, Raleigh, NC — 919-457-7841 — rsiddiq2@ncsu.edu —
linkedin.com/in/rashiksiddiquee — GitHub

Career Summary

- Ph.D. researcher specializing in Data Science, Geospatial Analytics, Machine Learning, and Optimization for large-scale decision-making.
- Experienced in building end-to-end analytics pipelines including data ingestion, feature engineering, modeling, interfaces and dashboard deployment.
- Strong background in predictive analytics, optimization modeling, and causal/statistical inference.
- Hands-on experience with Generative AI, LLM fine-tuning, LoRA adaptation, and Human-in-the-Loop data extraction systems.
- Hands-on experience with collaborating with research partners, including food banks, partner agencies
- Proficient in Python, SQL, R, Tableau, and modern ML frameworks.

Education

- **Ph.D. in Industrial Engineering**, North Carolina State University *Aug 2021 – Present*
  Major: Operations Research — Minor: Statistics
  CGPA: 3.85/4.00 — Expected Graduation: October 2026
  Supervisors: Dr. Julie Ivy and Dr. Osman Y. Özaltin
  Research Focus: Decision optimization and AI-driven analytics in food bank supply chains.
- **MBA in Finance**, Institute of Business Administration, University of Dhaka *May 2021*
  CGPA: 3.45/4.00
- **B.S. in Mechanical Engineering**, Bangladesh University of Engineering and Technology (BUET) 3.66/4.00 *Oct 2018*

Relevant Coursework

Dynamic Stochastic Optimization, Integer Programming, Linear and Non-linear Optimization, Network Flows, Logistics Engineering, Applied Stochastic Models, Design of Experiments, Time Series Analysis, Deep Learning, Dynamic Programming, Statistical Inference, Experimental Stat

Technical Skills

Programming Languages:	Python, R
Statistical Analysis Tools:	RStudio, SPSS, JMP, SAS
Visualization Tools:	PowerBI, Tableau, ArcGIS Pro
Database:	Microsoft SQL Server,PostgreSQL
Data Science and ML Tools:	SciKit-Learn, PyTorch, TensorFlow, Pandas, NumPy
Machine learning Skills:	Classification, Clustering, Random Forest, Forecasting
AI skills:	Prompt Engineering, LoRA, Finetuning, LMStudio, Huggingface
Statistical Techniques:	Regression Analysis, Causal Inference, Hypothesis Testing
Deployment Skils:	CI/CD pipeline, Git, Google Cloud Services, Virtual Machine

Experience

Graduate Teaching and Research Assistant,
Department of Industrial and Systems Engineering, NC State University *May 21 - Present*

- Collaborated with multiple food banks to access and understand their operational challenges
- Designed and led research initiatives including problem identification, method design and implementation
- Completed multiple projects and delivered to the stakeholder with regular maintenance

Business Intelligence Analyst,
UCB Fintech Company Limited, Dhaka, Bangladesh *Mar 21 - Aug 21*

- Designed data pipelines and integrated SQL-based data extraction processes to ensure scalability and efficiency.
- Developed predictive analytics models to enhance business decision-making, revenue forecasting, and support marketing target performance.
- Created interactive dashboards in Tableau for data-driven decision-making
- Implemented a data management-failure monitoring system that reduced failure rate from 10% to 3%.

Projects

- **Predicting willingness to collaborate along agencies:** *2023-Present*
  - Collected, curated and performed feature engineering on food bank operations data and collaborate history
  - Developed logistic regression, ML models including Random Forest, Bagging, Boosting methods to predict the willingness to collaborate among agencies
- **Equitable Accessibility Dashboard for Food Bank Agencies:** *2024-Present*
  - By creating a large composite transportation dataset, developed a dashboard in Tableau, to map public transport accessibility for food bank locations using GTFS data feed
  - Identified under-served and food-insecure populations and provided food bank partners with actionable insights. Link
  - Devised a mixed integer linear model to optimize the location of new agencies
- **Generative AI Data Extraction and Standardization for Food Bank Operations** *2024–Present*
  - Developed an end-to-end LLM pipeline using Ollama (LLaMA-3) and LM Studio to standardize unstructured agency operational data into JSON.
  - Integrated web-scraping with a Human-in-the-Loop (HIL) framework for data extraction, validation, achieving 60%+ extraction accuracy.
  - Performed a LoRA finetuning to improve raw text parsing accuracy to 93%.
  - Deployed the model to continuously process 800+ agency records for data integration and dashboard reporting with statistical process control.
- **Geo spatial tool to locate food support sources for Wake county residents** *2025-Present*
  - Developed an online food source locator tool that can help users find agencies based on user location
  - Deployed in multiple agencies as part of County-wide food security initiative Link
- **Clustering agencies for improved resilience:** *2023*
  - Designed an optimization model to cluster food bank agencies based on willingness to collaborate
  - Creating a bi-level optimization problem to capture the game theoretic nature of the problem.
- **Sustainability Practices in Food Rescue:** *2022-23*
  - Developed a multi-attribute performance analysis tool for evaluating the sustainable practices of organization in food supply chain industry
  - Applied machine learning algorithms (decision tree, XGBoost) to determine sustainability scores for nonprofit and for-profit organizations
- **Fraud Detection Model for Fintech Payments:** *2021*
  - Applied anomaly detection techniques like data pattern analysis and predictive modeling to identify at-risk transactions and prevent financial fraud.
  - Visualized data in Tableau to highlight fraud patterns and transaction trends
- **Fintech platform transaction failure analysis:** *2021*
  - Performed extensive data analysis of transaction data from a system perspective to determine the root cause of the high failure rate.

– Developed Pandas-SQL-based tool to validate data tables and streamline the large volume data flow for finance and supply chain team.
– Created a Tableau dashboard for management to demonstrate the system faults by end-to-end data lineage exploration.

Publication

- Siddiquee, R, Ivy, J. S., Ozaltin, O. Y. (2027).FoodBridge: Connecting Communities through Automated Information Extraction from Partner Websites, *Submitted for AAAI 2027*
- Siddiquee, R., Ivy, J. S., *et al.* (2025). A Framework for Characterizing and Evaluating Food Rescue and Waste Reduction Organizations. Annals of Op. Res *Manuscript submitted for publication.*
- Chowdhury, N. R., Siddiquee, R. I., and Ivy, J. S. (2025). Influence of Norms in Alliance Characteristics of Humanitarian Food Agencies: Capability, Compatibility and Satisfaction. Proceedings of the 2025 Winter Simulation Conference
- Siddiquee, R. I., Ivy, J. S., and Ozaltin, O. Y. (2025). Leveraging Generative AI for Structured Data Extraction and Processing in Food Bank Operations. INFORMS Data Mining and Decision Analytics (DMDA) Workshop 2025.

Honors and Awards

- Edward P. Fitts Fellowship *2023-24*
- Fitts Supplemental Fellowship *2021*
- INFORMS, Minority Issues Forum Webinar Finalist *2025*
- INFORMS Minority Issues Forum Travel Award *2026*

Conference Presentations

- **Food Rescue Landscape Analysis Framework to Characterize For-Profit and Nonprofit Organizations**
  INFORMS Annual Meeting 2023, 15-18 October 2023 *Phoenix, Arizona*
- **Designing a Resilient Food Bank Network through Information Sharing**
  Annual IISE Conference and Expo, 18-21 May 2024 *Montreal, Canada*
- **Exploring Equity in Accessibility in Food Bank Network**
  INFORMS Annual Meeting 2024, 20-23 October 2024 *Seattle, Washington*
- **Bridging the Gap: Quantifying Spatial and Temporal Access within a Food Bank Network**
  INFORMS Annual Meeting 2025, 26-29 October 2025 *Atlanta, Georgia*
- **Generative AI based Data Extraction for Food Bank Agencies**
  INFORMS Annual Meeting 2025, 26-29 October 2025 *Atlanta, Georgia*
- **Prediction Willingness to Collaborate in Food Bank Agencies**
  INFORMS Annual Meeting 2025, 26-29 October 2025 *Atlanta, Georgia*
- **Influence of Norms on Alliance Characteristics of Humanitarian Food Agencies: Capability, Compatibility, and Satisfaction**
  Winter Simulation Conference 2025, 7-10 December 2025 *Seattle, Washington*

Membership and Activities

- President, INFORMS NCSU Student Chapter
- Voluntary GIS and database consultant, Capital Area Food Network
- Member, Pack Essentials Advisory Board, NCSU

Mentorship

Sauem Jung-(Hanyang University, South Korea /Study Abroad student, Computer Science, NCSU)
Mundhir Al Maskari (NC State University - Sophomore ISE)